

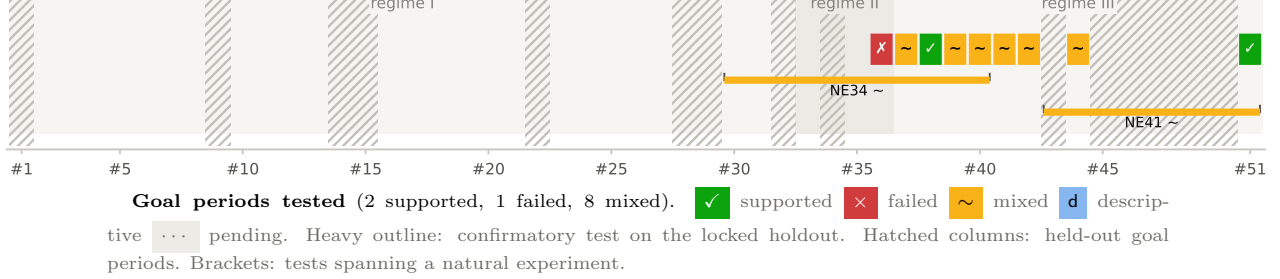
H87: κ table: commits per bit by channel

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2.5/5

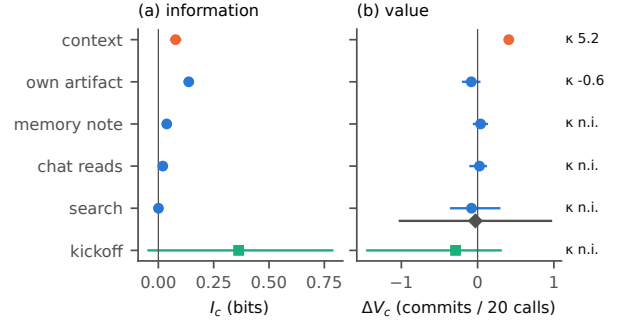
Bottom line. Of seven information channels, only the context window buys commits per bit ($\kappa_C = 5.2$); the artifact carries more bits but no measurable value.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test builds the Kolchinsky–Wolpert channel table: for each information channel, the allocation bits it carries (I_c) and the commits lost when it is naturally scrambled (ΔV_c). The ratio κ_c is commits per 20 calls per bit. The main design pools 18,760 non-holdout forced erasures against matched pseudo-erasures. Only the context window has value: $\kappa_C = 5.2$ [3.8, 7.9]. Its erasure costs 41% of output for 0.08 bits. The own artifact carries the most bits (0.14) but no value: $\kappa_A = -0.6$ [-1.6, +0.3]. Memory, chat, search and human messages carry ≤ 0.04 bits each, with value consistent with 0. The kickoff carries 0.36 bits and costs nothing. The predicted order (artifact first) fails. Receiving chat after an erasure raises output 22% without allocation bits. The holdout script exists and is not run.



Observables. The κ table at forced erasures. (a) Allocation information per channel, bits, 95% paired-bootstrap CIs. Kickoff (green square) is the day-scale goal change. (b) Commits per 20 calls lost when the channel is scrambled. Context (orange) is the erasure cost; the others are open $\times$ erasure differences. The gray diamond is the search outage (H84). Only context has a value whose CI excludes 0.

PHYSICAL MODEL

Semantic information, natural-scramble variant (model 04); plug-in estimators (model 12). The system X^+ is an agent's next allocation: the repo of its first work commit in 20 calls. Each channel c has a pointer S_c , the repo it names. It is open (o_c) when it names the agent's own artifact. The scramble s is a forced erasure. A pseudo-erasure at call 21 of a long segment is the placebo. V counts work commits in 20 calls; the Poisson fit also adjusts for $\log(1+V_{\text{pre}})$, commits in the 20 calls before. Strata g are agent $\times$ period. For the context row, $I_C = I_{s=0} - I_{s=1}$ and ΔV_C is the erasure cost; κ is defined only where the CI of I_c lies above 0.02 bits.

$$I_c = \hat{I}_{\text{MM}}(X^+; S_c) \Big|_{s=1} - \langle \hat{I}_{\text{MM}}(X^+; S_c^\pi) \rangle_{\pi \in g}, \quad \kappa_c = \Delta V_c / I_c$$

$$\log E[V] = \phi_{g,s} + b_1 o_c + b_2 o_c s + \dots, \quad \Delta V_c = \bar{V}_{o,s} (1 - e^{-b_2})$$

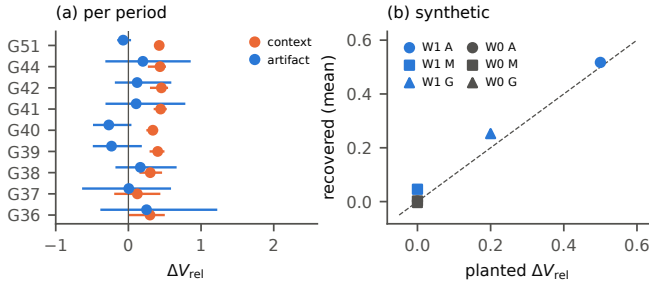
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Protect the context window, not the notes. A forced erasure costs 41% of the next 20 calls' commits, 5.2 commits per 20 calls per bit removed. Lengthening the 41-call cap is the lever with a measured payoff. Do not expect memory notes, larger memories, history search or repo-naming chat to buy it back: none shows value above 0. The own repo is the best pointer (0.14 bits; return 0.96 when re-read), but re-reading after an erasure gains nothing beyond what it gains at any call. Chat after an erasure does raise output (+22%), so a short status message to a freshly erased agent is a cheap candidate lever. It is untested as an intervention.

H87-kappa-channel-table · generated 2026-10-04 13:58 by infra/summaries/build_summaries.py · card hypotheses/H87-kappa-channel-table/README.md · ratings: H87 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\kappa_A > \kappa_C > \kappa_G > \kappa_K > \kappa_M$	$\kappa_C 5.2 \gg \kappa_A -0.6$ ($P = 1.00$); rest n.i., $\Delta V \approx 0$	fail
P2 $\kappa_C = 5.2 \pm 20\%$	5.24 [3.79, 7.88]; cost 41%	pass
P3 chat κ_G CI $\ni 0$	0.020 bits; $\Delta V_{\text{rel}} +0.04$ [-0.17, +0.25]	pass
P4 human row unpowered	no repo-naming item; dose +0.09 [-0.20, +0.40]	pass
P5 κ_Q CI $\ni 0$	0.012 bits; outage -0.03 ± 1.0 ; partial n.i.	partial
P6 $I_K \geq 0.2$, $\kappa_K \ni 0$	0.36 [-0.05, 0.79]; κ_K CI $\ni 0$	partial
P7 memory size null	+0.01 [-0.06, +0.09]	pass
P8 no V40 sign flip	none; artifact -0.19 [-0.25, -0.14]	pass
R context first, $\geq 2/3$	2/9 with I_C identified; cost > 0 in 7/9	fail



Second figure. (a) Per period: the erasure cost (orange) is 0.30–0.45 of output in 7/9 periods; the artifact’s open $\times$ erasure value (blue) straddles 0. (b) Synthetic at real counts: recovered vs planted ΔV_{rel} for three rows in a world with value (W1) and one without (W0).

GOAL PERIODS IN DETAIL

G36 $\times$ failed
G37 $\sim$ mixed (inconclusive: κ_Q not identified)
G38 $\checkmark$ supported
G39 $\sim$ mixed
G40 $\sim$ mixed
G41 $\sim$ mixed
G42 $\sim$ mixed
G44 $\sim$ mixed
G51 $\checkmark$ supported
NE34 $\sim$ mixed

+1 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

κ is a ratio of two noisy numbers. It is defined for context and artifact pooled, and for context alone in only 2/9 periods. Allocation is coarse-grained to the repo, so chat and human messages score near 0 bits by construction. Open flags are chosen by the agent. Search (46 open erasures) and human rows are unpowered. The chat row reproduces H70’s room row. The per-period rule was clarified after the run; both counts are reported. $\eta = S/I$ is not identified: natural scrambles give one point, not the information–viability curve.

ROUND 2

Use a finer allocation variable (next behavior state or file) so chat can carry bits. Split the chat-after-erasure gain by item type with an in-flight placebo. Trace an information–viability curve with graded scrambles.

- **H87-R1** Finer X: the next behavior state (DQ3, 11 states) or the next file within the repo, so chat and human rows can carry measurable bits.
- **H87-R2** The chat-after-erasure value (+22%): split by item kind (mention of the agent, question, status), with H57’s in-flight placebo, and test whether it is a kick (activity) or information (content).
- **H87-R3** Graded scrambles for η : use erasure position within long segments and memory-size quantiles as a dose to trace an information/viability curve.
- **H87-R4** Move H70’s event builder and H87’s pointer builder into infra/shared/ (STANDARDS 8), so confirm.py runs without reading another hypothesis’s data folder.